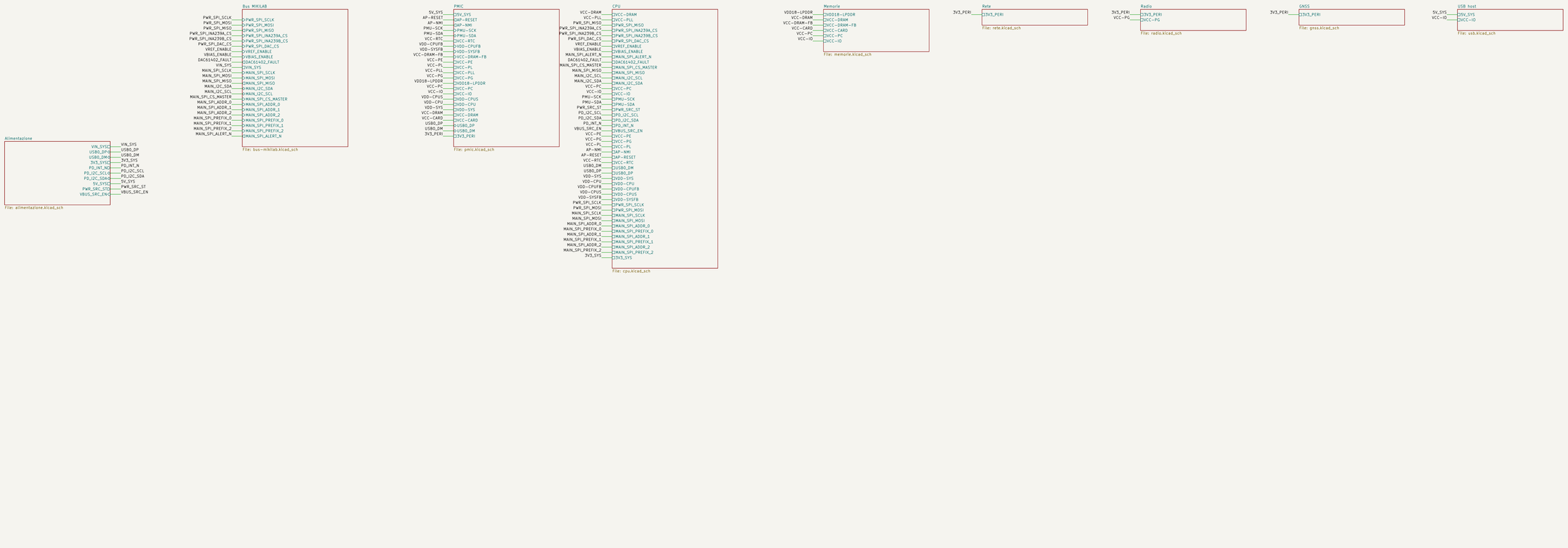


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CPU: Allwinner A133 (LFBGA-346). Ball map dal reference Allwinner A100_A133_B3_AXP707_LPDDR3 (pagine 06-08). Serie componenti 3xx.

Porte verso i fogli futuri: DRAM_* (LPDDR4), EMMC_* (PC), SD_* (PF), WL_*/BT_* (AP6256 su PG + PL4/PL5). RGMII_*/PHY_* (RTL8211F su PH). GNSS_* (UART3 PD14/PD15, PPS su PG10).

Bus: interfaccia-bus-mkllab rev. 4, FUSB302B su S_TWI1 (PL8/PL9) con pull-up a VCC-PL: 3V3_SYS del FUSB302B preso da VCC-PL tramite R 0R.

Clock come nel reference: quarzo 24 MHz su DXIN/DXOUT (18 pF), 32.768 kHz su X32KIN/X32KOUT (22 pF, 10M in parallelo). WREQIN a GND (fanout 24 MHz non usato).

Reset/FEL AP-RESET dal PWROK del PMIC + tasto RESET; tasto FEL (BOOT) a GND. Strap GPADC0 = 10k/3k9 (riga 4: SMHC0 poi eMMC). GPADC1 DDR PARA come nel reference (DNP).

Disaccoppiamento come nel reference: VDD-CPU 10uF + 2x2.2uF. VDD-SYS 10uF + 2x2.2uF. 0.1uF su VDD-CPUS/VDD-USB/VCC-EFUSE/VCC-RTC/VCC-PLL/VCC-USB. VCC-IO 1uF + 0.1uF.

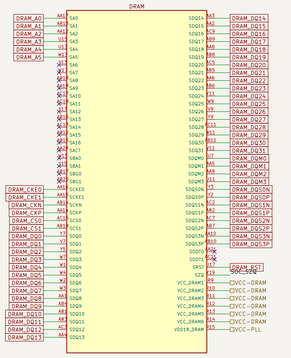
Analogici: AVCC 2.2uF. VRA1/VRA2 470nF verso AGND. AGND a GND con 0R. CPVDN 10uF. VEE/CPVEE 2.2uF (stella, non fly-by). CPVDD 2.2uF.

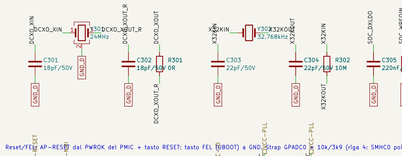
Bus: 33R serie vicino al SoC su CLK/MOSI di SPI2 (BUS_PWER) e SPI1 (MAIN); 100R serie su RGOR/PREFIX MISO dirette.

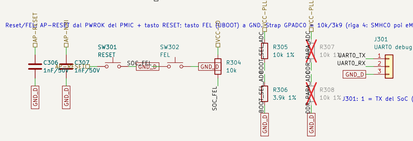
Livelli di riposo sicuri: CS e CS_MASTER alti (R0A); VREF_ENABLE alto = riferimento spento (attivo basso, 10k); VBIAS_ENABLE basso = bias spento (100k a GND).

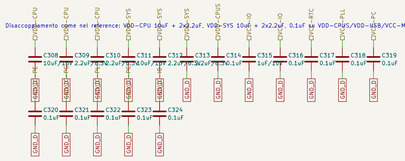
ALERT# A 3V3 unico pull-up del bus. DAC61402_FAULT 10k (open-drain). I2C0 del bus 2.2k. S_TWI1 del FUSB302B 2.2k a VCC-PL. PD_INT_N 10k.

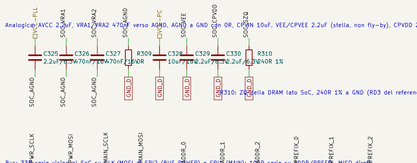
ESD bassa capacità TPD4E05U06DQA (0.5 pf, 5.5 V) sulle linee logiche di J1/J2, da piazzare vicino ai connettori. Nessuna protezione sui V_BIAS (30 V) e sui riferimenti.

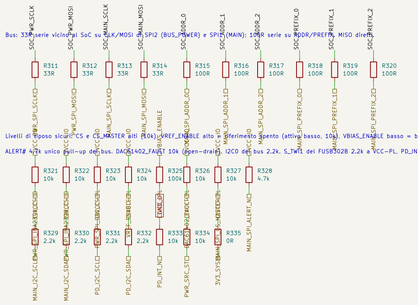


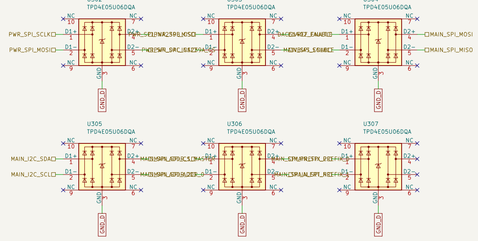


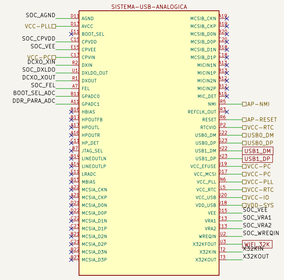


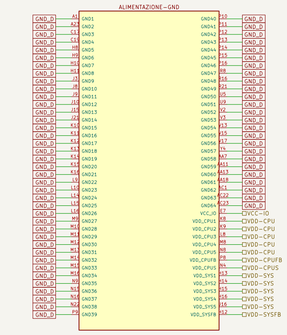


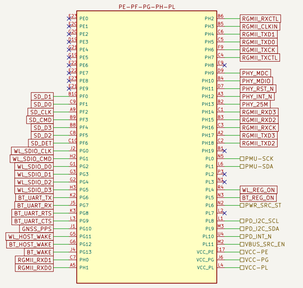


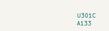












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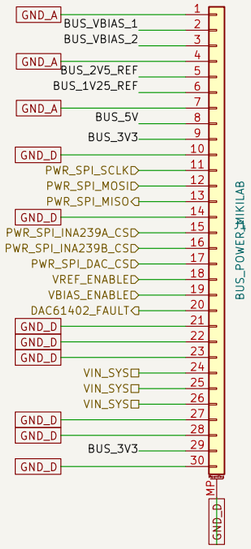
J1 BUS_POWER 30 pin (Hirose FH12-30S): pin 1-22 come J3 del modulo Power, 24-26 VIN_SYS 12 V, 29 secondo 3V3. Rail BUS_* passanti verso J2, non collegati ai rail della scheda.

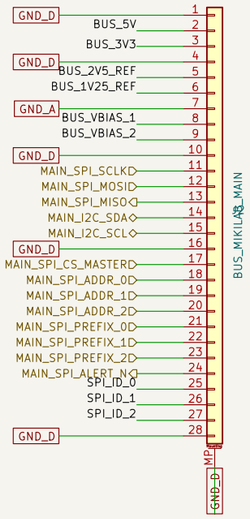
VIN_SYS -> buck TPS563201 (12->5 V, 3 A) -> ideal diode LM66100 -> nodo jack DC a monte del TPS2121 (foglio alimentazione). BUS_5V va solo a J2 pin 2.

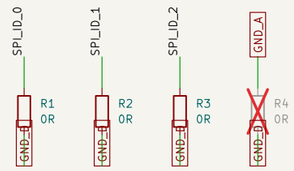
J2 MAIN 28 pin secondo BPS-MIKILAB-2026-REV5.0 (spec: XUNPU FPC-05F-28PH20; footprint Hirose FH12-28S provvisorio, da verificare).

Da aggiungere sul foglio CPU: 33R serie su SPI, 100R su ADDR/PREFIX, pull-up 4.7k su ALERT, 2.2k su I2C, 10k sui CS, ESD TPD4E05U06.

R4 0R: ponte GND_A-GND_D, DNP di default (sul modulo Power GND_A e GND_D sono gia la stessa net). R1-R3: ID iniziale 000 (scheda = J_OUT, testa della catena).







Sheet: /Bus MIKILAB/
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1) VIN_SYS 12 V dal BUS_POWER (pin 24–26) -> F101 2.5 A -> buck TPS56637 12 -> 5 V / 6 A -> ideal diode LM74700 + CSD17551Q5A -> 5V_EXT

2) Sorgente DC (barrel 5.5×2.1 mm o header) -> F102 4 A -> ideale diode -> 5V_EXT. Divisore R105/R106 = 10k/10k = 4.99 V, L 3.3 uH, Cout 4x22 uF, feed-forward R116 20k + C126 100 pF.

J103
DC-005-5.5×2.0/2.1
EN: 169k/36.1k (schema tipico TI). MODE = FCCM, R114 DNP per Eco-mode. Q101 N-MOSFET 30 V, Vgs +-20 V (il LM74700 pilota fino a 13 V).

4) Power mux TPS2121: IN1 5V_EXT (jack o VIN_SYS) con priorita, IN2 VBUS USB-C. Uscita 5V_SYS verso AXP707 (VBUS/ACIN)

PR1: IN1 sotto 4.49 V -> passa a IN2. OV1/OV2: canale escluso sopra 5.98 V. CP2 a GND (modo VREF). Ilm 22.1k -> 4–5 A. SS 100 nF.

5) SY6280AAC: VBUS in uscita verso una periferica (source, -1 A). Ingresso da 5V_EXT: si accende solo con jack o VIN_SYS presenti

Ilim = 6800 / R112 = 1 A. R113: uscita spenta finche Linux non abilita VBUS_SRC_EN (comandato da tcpm).

3) USB-C (HRO TYPE-C-31-M-12): solo USB 2.0, FUSB302B per CC/PD, fusibile 3 A e TVS su VBUS

FUSB302B: pull-up I2C 2.2k e INT_N 10k sul foglio CPU (I2C condivisa). VCONN non usato: verificare sul datasheet se lasciarlo libero.

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PMIC AXP717 secondo il reference Allwinner A100_A133_B3_AXP707_LPDDR3 (pag. 05, che usa l'AXP717). Ingresso 5V_SYS dal TPS2121. Niente batteria.

Net rinominate come nel reference. Alias: VCC-PLL = AVCC/VCC-PLL/VDD18-DRAM; VCC-PC = VCC-PC/VCC-EFUSE/CPVIN/VCC-LVDS (1.8 V, VCCQ eMMC);

VCC-IO = VCC-IO/VCC-USB/VCC-PD/VCC-3V3 (3.3 V); VCC-CARD = VCC-CARD/VCC-NAND (microSD + VCC eMMC). LPDDR4: VDD2/VDDQ = VCC-DRAM 1.1 V, VDD1 = VDD18-LPDDR 1.8 V. Niente VPP (solo DDR4).

Ingresso VBUS = 5V_SYS, VMID, induttore del power path SW = PMU_SW, VSYS (PS), BAT senza batteria: solo condensatori, caricatore da disabilitare in U-Boot.

C201 10uF/10V · C202 22uF/10V · L201 1.0uH/3A · C203 22uF/10V · C204 22uF/10V · C205 4.7uF/10V · C206 100nF · R201 10k 1%

R201 su TS = NTC fittizio a 25 C, come RP45 del reference.

Condensatori degli ingressi VIN1-3 / ALDOIN / BLDOIN / CLDOIN vicini al PMIC. VREF, VRTC, CPUSLDO

C207 1uF/10V · C208 1uF/10V · C209 1uF/10V · C210 1uF/10V · C211 1uF/10V · C212 1uF/10V · C213 1uF/10V · C214 1uF/10V · C215 2.2uF/6.3V

DCDC1 VDD-CPU 0.9-1.18 V 4 A, DCDC2 VDD-SYS/VDD-USB 0.95 V, DCDC3 VCC-DRAM 1.1 V (LPDDR4). FB con sense remoto: FB1/FB2 dalle sfere dei BGA, FB3 vicino alla DRAM.

L202 1.0uH/3A · C216 22uF/10V · C217 22uF/10V · L203 1.0uH/3A · C218 22uF/10V · C219 22uF/10V · L204 1.0uH/3A · C220 22uF/10V · C221 22uF/10V

LDO: ALDO1 VCC-PE 1.8 V, ALDO3 VCC-PLL 3.3 V, ALDO4 VCC-PLL 1.8 V, BLDO1 VCC-PG 3.3 V (VCCIO-WIFI), BLDO2 VDD18-LPDDR 1.8 V, CLDO1 VCC-PC 1.8 V, CLDO3 VCC-IO 3.3 V.

Non usati (camera, LCD, touch, motore): ALDO1, BLDO3, BLDO4, CLDO2, CLDO4. Condensatore montato, uscita spenta da software.

C222 2.2uF/6.3V · C223 2.2uF/6.3V · C224 2.2uF/6.3V · C225 2.2uF/6.3V · C226 2.2uF/6.3V · C227 2.2uF/6.3V · C228 2.2uF/6.3V · C229 2.2uF/6.3V · C230 2.2uF/6.3V · C231 2.2uF/6.3V · C232 2.2uF/6.3V · C233 2.2uF/6.3V

LDO esterno 3.3 V per microSD ed eMMC (come U19 TCS2155-E33 del reference, qui AP2112K-3.3 stesso pinout SOT-23-5). EN da VCC-IO: parte dopo CLDO3.

U202 AP2112K-3.3 · C234 1uF/10V · C235 1uF/10V · C236 10uF/10V

U201 AXP717

5V_SYS	49 VBUS_1	VIN1 9	PS
5V_SYS	50 VBUS_2	LX1_1 10	LX1
VMID	47 VMID_1	LX1_2 11	LX1
VMID	48 VMID_2	FB1 12	VDD-CPUFB
PMU_SW	45 SW_1	VIN2 32	PS
PMU_SW	46 SW_2	LX2 33	LX2
PS	43 VSYS_1	FB2 34	VDD-SYSFB
PS	44 VSYS_2	VIN3 31	PS
VBAT	41 BAT_1	LX3 30	LX3
VBAT	42 BAT_2	FB3 29	VCC-DRAM-FB
PMU_TS	39 TS	ALDOIN 15	PS
	38 CHGLED	ALDO1 13	ALDO1_NU
BACKUP	36 BACKUP	ALDO2 14	VCC-PE
PWRON	6 PWRON	ALDO3 16	VCC-PL
AP-RESET	1 PWROK	ALDO4 17	VCC-PLL
AP-NMI	37 IRQ	BLDOIN 20	PS
PMU-SCKD	52 SCK	BLDO1 18	VCC-PG
PMU-SDAD	51 SDA	BLDO2 19	VDD18-LPDDR
PMU_DP	3 DP	BLDO3 21	BLDO3_NU
PMU_DM	2 DM	BLDO4 22	BLDO4_NU
	4 CC1	CLDOIN 25	PS
	5 CC2	CLDO1 23	VCC-PC
DCDCEN	40 DCDCEN	CLDO2 24	CLDO2_NU
PMU_VREF	7 VREF	CLDO3 26	VCC-IO
VCC-RTC	35 VRTC	CLDO4 27	CLDO4_NU
	GND EPAD 53	CPUSLDO 28	VDD-CPUS

Controllo verso l'A133: PMU-SCK/PMU-SDA = bus RSB (S_RSB), AP-NMI = NMI, AP-RESET = PWROK verso RESET del SoC, PWRON = tasto di accensione.

SW201 PWRON · C237 100nF · R202 10k 1% · R203 10k 1% · R204 2k · R205 2k · R206 470R · R207 470R 1%

R204/R205 DNP: RSB non vuole pull-up (NO anche nel reference). R206/R207 DNP: rilevamento BC1.2 sul USB0, inutile con il FUSB302B.

RTC: pila CR1220 opzionale DNP su BACKUP, senza pila l'ora va presa dal GNSS/NTP.

BT201 CR1220 (opt.) · R208 0R · C238 100nF

3V3_PERI: TPS563201 da 5V_SYS, 3.3 V / 3 A per AP6256 (VBAT), RTL8211FS e NEO-F10T. Separato dalla 3V3 del bus. EN da DCDCEN del PMIC (stessa sequenza

U203 TPS563201 · C239 10uF/10V · C240 100nF · C241 100nF · L205 2.2uH/4A · R209 0R · R210 100k · R211 33.2k 1% · R212 10k 1% · C242 22uF/10V · C243 22uF/10V

Vout = 0.768 V x (1 + R211/R212) = 3.32 V. R210 DNP: alternativa per accenderlo sempre da 5V_SYS se DCDCEN non va bene.

Sheet: /PMIC/	
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MEMORIE: LPDDR4 2 GB x32, eMMC 256 GB su SMHC2 (PC), microSD su SMHC0 (PF). Serie componenti 4xx.

Le net DRAM_*, EMMC_*, SD_* sono etichette globali condivise con il foglio CPU (porte tolte dal blocco CPU).

U401 LPDDR4 MT53E512M32D1ZW-046 IT:B (2 GB, 1 die, 2 canali x16). Collegamento come nel reference Allwinner A100_A133 (pag. 08, colonna LPDDR4):

CA0-5 <- SA5,SA0,SA2,SA1,SCS0,SCS1 del SoC; CS0/CS1 <- SA4/SA3: byte scambiati (canale A = DQ8-15 e DQ0-7, canale B = DQ16-23 e DQ24-31) come nel reference.

ODT_CA_A/B a VDD2 con 100R; ZQ0/ZQ1 240R 1% a VDDQ (VCC-DRAM), come nel reference. CS1/CKE1 vanno su ball NC del chip a singolo rank: innocui.

Disaccoppiamento vicino alla LPDDR4: VDD2/VDDQ 4x1uF + 6x0.1uF. VDD1 2x1uF + 2x0.1uF (reference + margine: da rifinire con la PDN Micron).

C415 vicino al SoC (C211 del reference).

R423: punto di sense FB3 del PMIC preso sulle sfere VDD2 della LPDDR4 (nota del reference).

U402 eMMC Kingston EMMC256-IY29-5B101 (HS200): VCC = VCC-CARD 3.3V, VCCQ = VCC-PC 1.8 V. CLK 33R serie vicino al SoC, pull-up 10k su CMD e DAT0-7 a VCCQ. RST_n 10k.

J401 microSD (Hirose DM3AT con switch di rilevamento) su SMHC0: VDD = VCC-CARD, CLK 33R serie, pull-up 47k su CMD/DAT a VCC-CARD, SD_DET su PF6 (10k a VCC-IO).

ESD TPD4E05U06DQA vicino allo slot: CLK (lato slot), CMD, DAT0-3, DET.

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RETE: PHY Realtek RTL8211F-CG (datasheet rev. 1.7) su RGMII banco PH (VCC-IO 3.3 V). Serie 5xx. Net RGMII_*/PHY_* globali verso il foglio CPU.

Strap (4.7k): PHYAD = 001, PLLOFF = 0, TXDLY = 1 e RXDLY = 1 (rgmii-id, ritardi nel PHY), CFG_EXT = 1 con CFG_LDO = 00: pad RGMII alimentati da fuori a 3.3 V.

U501
RTL8211F-CG

PHY_TXC_P 20 TXC
PHY_TXD0_P 18 TXD0
PHY_TXD1_P 17 TXD1
PHY_TXD2_P 16 TXD2
PHY_TXD3_P 15 TXD3
PHY_TXCTL_P 19 TXCTL
RGMII_RXCK 27 RXC/PHYAD1
RGMII_RXD0 25 RXD0/RXDLY
RGMII_RXD1 24 RXD1/TXDLY
RGMII_RXD2 23 RXD2/PLLOFF
RGMII_RXD3 22 RXD3/PHYAD0
RGMII_RXCTL 26 RXCTL/PHYAD2
PHY_MDC 13 MDC
PHY_MDIO 14 MDIO
PHY_INT_N 31 INTB/PMEB
PHY_RST_N 12 PHYRSTB
PHY_CLKOUT 35 CLKOUT
PHY_XI 36 XTAL_IN
PHY_XO 37 XTAL_OUT

1 MDI0P MDI0_P
2 MDI0N MDI0_N
4 MDI1P MDI1_P
5 MDI1N MDI1_N
6 MDI2P MDI2_P
7 MDI2N MDI2_N
9 MDI3P MDI3_P
10 MDI3N MDI3_N
32 LED0/CFG_EXT PHY_LED0
33 LED1/CFG_LDO0 PHY_LED1
34 LED2/CFG_LDO1 PHY_LED2
39 RSET PHY_RSET
30 REG_OUT PHY_REG_OUT
21 DVDD10 PHY_VDD10
3 AVDD10 PHY_VDD10
8 AVDD10 PHY_VDD10
38 AVDD10 PHY_VDD10
11 AVDD33 3V3_PERI
40 AVDD33 3V3_PERI
29 DVDD33 3V3_PERI
28 DVDD_RG 3V3_PERI
41 EPAD_GND GND_D

22R serie vicino al SoC su TXC, TXD0-3, TXCTL (MAC -> PHY). RX senza serie (sorgente PHY).

R501 22R RGMII_TXCK -> PHY_TXC_P
R502 22R RGMII_TXD0 -> PHY_TXD0_P
R503 22R RGMII_TXD1 -> PHY_TXD1_P
R504 22R RGMII_TXD2 -> PHY_TXD2_P
R505 22R RGMII_TXD3 -> PHY_TXD3_P
R506 22R RGMII_TXCTL -> PHY_TXCTL_P

Strap su RXD/RXC/RXCTL e LED (4.7k). LED1/LED2 attivi alti verso i LED del RJ45 (+10R); LED0 solo strap CFG_EXT.

R507 4.7k 3V3_PERI -> RGMII_RXD3
R508 4.7k RGMII_RXCK -> GND_D
R509 4.7k RGMII_RXCTL -> GND_D
R510 4.7k RGMII_RXD2 -> GND_D
R511 4.7k 3V3_PERI -> RGMII_RXD1
R512 4.7k 3V3_PERI -> RGMII_RXD0
R513 4.7k 3V3_PERI -> PHY_LED0
R514 4.7k PHY_LED1 -> GND_D
R515 4.7k PHY_LED2 -> GND_D

MDIO 1.5k a 3.3 V, INT 4.7k, PHYRSTB 10k (il SoC tiene il reset per almeno 10 ms dopo l'accensione). RSET 2.49k 1%.

R516 1.5k 3V3_PERI -> PHY_MDIO
R517 4.7k 3V3_PERI -> PHY_INT_N
R518 10k 3V3_PERI -> PHY_RST_N
C501 100nF PHY_RST_N -> GND_D
R519 2.49k 1% PHY_RSET -> GND_D
R520 0R PHY_CLKOUT -> RGMII_CLKIN
R521 0R PHY_25M -> PHY_XO

R520/R521 DNP: CLKOUT verso PHY e 25 MHz da PH13 (EPHY0_25) come alternativa al quarzo, non usati di default.

Quarzo 25 MHz +-50 ppm (requisito datasheet) con 2x 15 pF. Switching regulator 1.0 V: REG_OUT -> 2.2 uH -> AVDD10/DVDD10.

PHY_XI -> Y501 1 25MHz 2 -> PHY_XO
C502 15pF PHY_XI -> GND_D
C503 15pF PHY_XO -> GND_D
L501 2.2uH PHY_REG_OUT -> PHY_VDD10
C504 22uF PHY_VDD10 -> GND_D
C505 0.1uF PHY_VDD10 -> GND_D
C506 0.1uF PHY_VDD10 -> GND_D
C507 0.1uF PHY_VDD10 -> GND_D
C508 0.1uF PHY_VDD10 -> GND_D
C509 10uF 3V3_PERI -> GND_D
C510 0.1uF 3V3_PERI -> GND_D
C511 0.1uF 3V3_PERI -> GND_D
C512 0.1uF 3V3_PERI -> GND_D
C513 0.1uF 3V3_PERI -> GND_D

RJ45 HanRun HR911130A (100/1000Base-T, magnetici e LED integrati). Prese centrali lato PHY (P1, comuni) a GND con 100 nF. P10 = rete Bob Smith verso lo schermo.

J501
HR911130A

MDI0_P P2 MDI0+
MDI0_N P3 MDI0-
MDI1_P P4 MDI1+
MDI1_N P7 MDI1-
MDI2_P P5 MDI2+
MDI2_N P6 MDI2-
MDI3_P P8 MDI3+
MDI3_N P9 MDI3-
ETH_CT P1 CT

11 LED_G_A ETH_LED1_A
12 LED_G_K GND_D
14 LED_Y_A ETH_LED2_A
13 LED_Y_K GND_D
P10 BOB_SMITH GND_D
SHIELD0 GND_D
SHIELD1 GND_D

ETH_CT -> C514 100nF -> GND_D

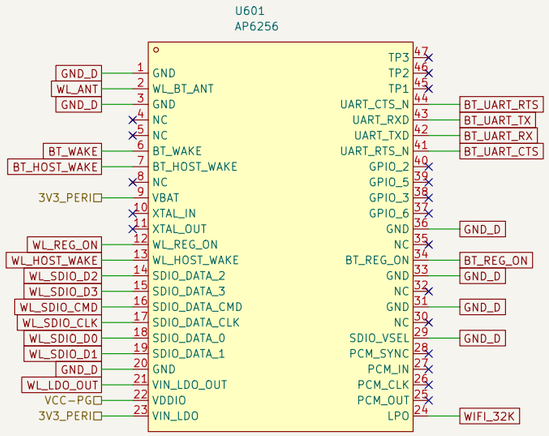
PHY_LED1 -> R522 510R -> ETH_LED1_A
PHY_LED2 -> R523 510R -> ETH_LED2_A

LED verde = LED1 del PHY (100M), giallo = LED2 (1000M), entrambi attivi alti.

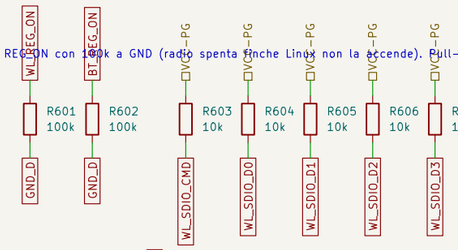
Sheet: /Rete/	
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Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 7/10

RADIO: AMPAK AP6256 (Wi-Fi 5 + BT 5.0) su SMHC1/PG (SDIO), UART1 con RTS/CTS, REG_ON su PL4/PL5. Serie 6xx. Net WL_*/BT_*/WIFI_32K globali.

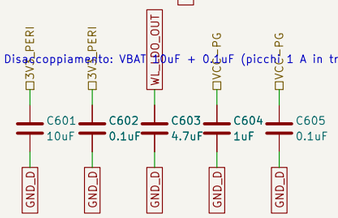
VDDIO = VCC-PG 3.3 V, quindi SDIO_VSEL a GND (SDIO 2.0, fino a 50 MHz). VBAT e VIN_LDO da 3V3_PERI. LPO 32.768 kHz da X32KFOUT.



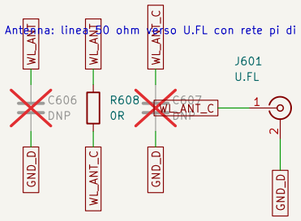
REG_ON con 100k a GND (radio spenta finché Linux non la accende). Pull-up 10k su CMD e DAT0-3 a VCC-PG.



Disaccoppiamento: VBAT 10uF + 0.1uF (picchi 1 A in trasmissione), VIN_LDO_OUT 4.7uF, VDDIO 1uF + 0.1uF.



Antenna: linea 50 ohm verso U.FL con rete pi di adattamento (OR + 2x DNP) da tarare.



UART incrociata: TXD del modulo -> BT_UART_RX (PG7), RXD <- BT_UART_TX (PG6), RTS_N -> BT_UART_CTS (PG9), CTS_N <- BT_UART_RTS (PG8).

PCM (25-28) non collegati: niente audio Bluetooth. XTAL_IN/OUT liberi (quarzo interno). GPIO e TP liberi.



Sheet: /Radio/
File: radio.kicad_sch

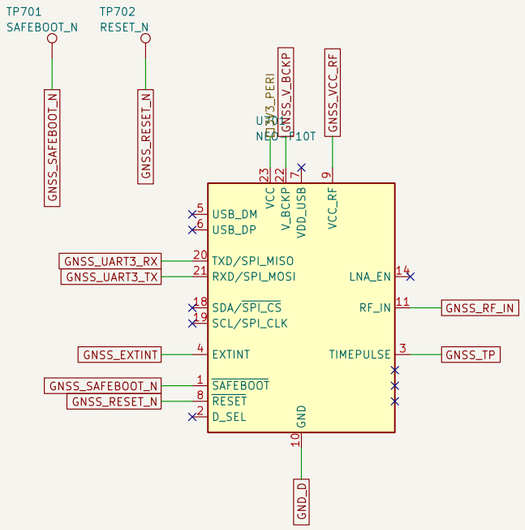
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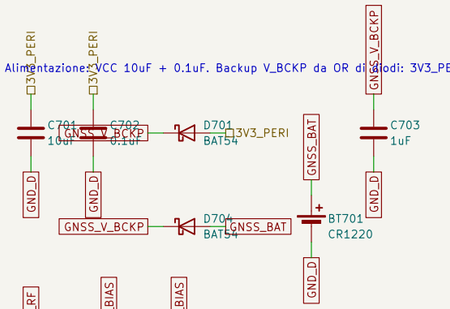
KiCad E.D.A. 10.0.4 | Id: 8/10

GNSS: u-blox NEO-F10T (timing L1/L5) in formato NEO (simbolo NEO-8Q di KiCad, pin da verificare sul datasheet F10T). Serie 7xx.

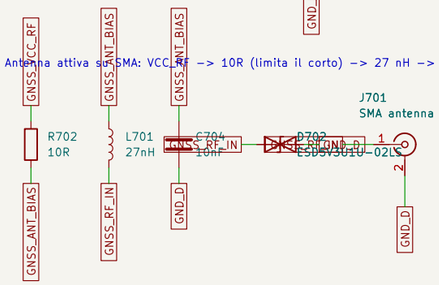
UART3: TXD del modulo -> GNSS_UART3_RX (PD15), RXD <- GNSS_UART3_TX (PD14). TIMEPULSE -> 33R -> GNSS_PPS (PG10, EINT).



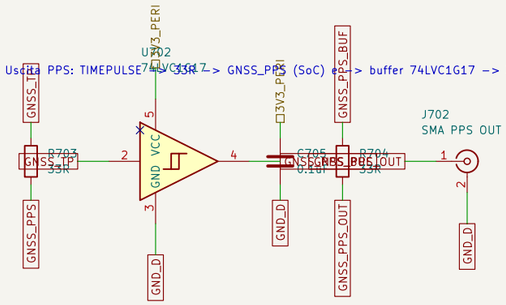
Alimentazione: VCC 10uF + 0.1uF. Backup V_BCKP da OR di diodi: 3V3_PERI (BAT54) e pila CR1220 (BAT54), nessuna ricarica della pila.



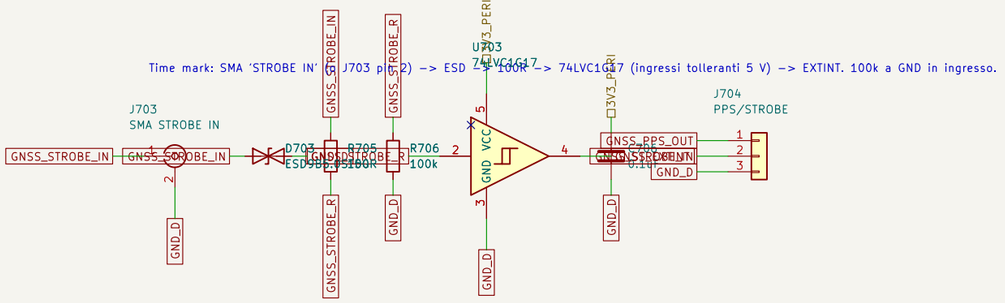
Antenna attiva SMA: VCC_RF -> 10R (limita il corto) -> 27 nH -> RF_IN. ESD a bassa capacita sul centrale. Linea 50 ohm corta.



Uscita PPS: TIMEPULSE -> 33R -> GNSS_PPS (SoC) e -> buffer 74LVC1G17 -> 33R -> SMA 'PPS OUT' e header J703. Nessun pull sul TIMEPULSE.



Time mark: SMA 'STROBE IN' -> J703 pin (3,2) -> ESD -> 100R -> 74LVC1G17 (ingressi tolleranti 5 V) -> EXTINT. 100k a GND in ingresso.



PWR_FLAG

GNSS_V_BCKP

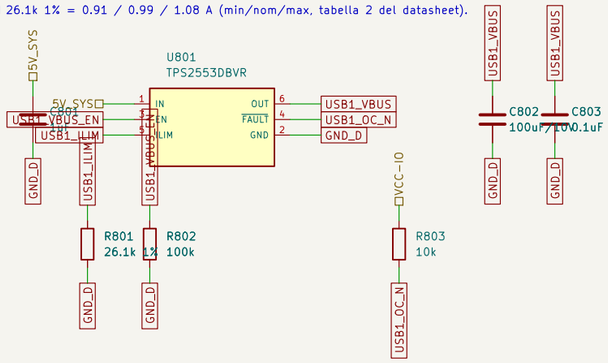
Sheet: /GNSS/		
File: gnss.kicad_sch		
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Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 9/10

USB HOST: USB1 del SoC su USB-C (16 pin, HRO TYPE-C-31-M-12), solo host. Serie 8xx. Net USB1_* globali verso il foglio CPU.

La porta non alimenta mai la scheda: VBUS esce solo dal load switch. Rp fissi da 22k a 5V_SYS su CC1/CC2 = sorgente 1.5 A (Type-C), niente controller PD.

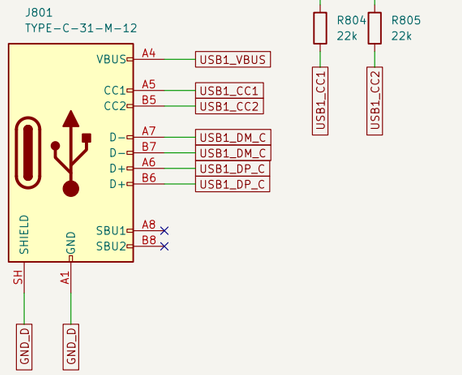
Load switch TPS2553 (ha il pin FAULT, che il SY6280AAC non ha): EN attivo alto da PD2 con 100k a GND, FAULT open-drain su PD3 con 10k a VCC-IO.

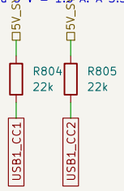
RILIM 26.1k 1% = 0.91 / 0.99 / 1.08 A (min/nom/max, tabella 2 del datasheet).



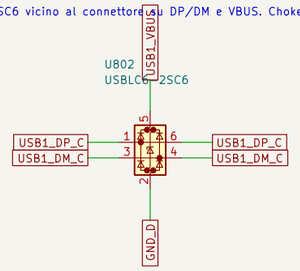
J801 USB-C: DP/DM collegati su entrambi i lati (A6/B6, A7/B7), SBU non connessi, schermo a GND.

Rp 22k a 5 V = 1.5 A. A 3.3 V per 1.5 A servirebbero 12k (10k a 3.3 V annuncia gia 3 A).

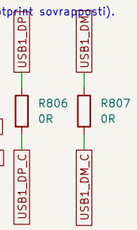




ESD USBLC6-2SC6 vicino al connettore su DP/DM e VBUS. Choke di modo comune DNP con 0R in parallelo (footprint sovrapposti).







Sheet: /USB host/		
File: usb.kicad_sch		
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